

Lesson 2.09.5: Solving Quadratic Equations Using the Quadratic Formula

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A **quadratic equation** is an equation that can be written in the standard form

$$ax^2 + bx + c = 0,$$

where $a \neq 0$.

Quadratic Formula

The **quadratic formula** is used to solve any quadratic equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Before Using the Formula

Before using the formula:

Write the equation in **standard form**.

Identify the values of a , b , and c correctly.

Substitute the values into the quadratic formula.

The Discriminant

The **discriminant** is

$$b^2 - 4ac.$$

It determines the nature of the roots.

Positive Discriminant

If

$$b^2 - 4ac > 0,$$

the equation has **two distinct real solutions**.

Zero Discriminant

If

$$b^2 - 4ac = 0,$$

the equation has **one repeated real solution.**

Negative Discriminant

If

$$b^2 - 4ac < 0,$$

the equation has **two complex solutions**.

Simplifying Solutions

Simplify radicals whenever possible.

Write the final answers in **simplest form**.

Always verify the solutions by substituting them into the original equation when necessary.

Examples

Example 1

Solve using the quadratic formula.

$$x^2 - 5x + 6 = 0$$

$$a = 1, \quad b = -5, \quad c = 6$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(6)}}{2(1)}$$

$$x = \frac{5 \pm \sqrt{1}}{2}$$

$$x = 3 \quad \text{or} \quad x = 2$$

Example 2

Solve using the quadratic formula.

$$2x^2 + 7x + 3 = 0$$

$$a = 2, \quad b = 7, \quad c = 3$$

$$x = \frac{-7 \pm \sqrt{7^2 - 4(2)(3)}}{2(2)}$$

$$x = \frac{-7 \pm 1}{4}$$

$$x = -\frac{3}{2} \quad \text{or} \quad x = -2$$

Example 3

Solve using the quadratic formula.

$$x^2 + 6x + 13 = 0$$

$$a = 1, \quad b = 6, \quad c = 13$$

$$x = \frac{-6 \pm \sqrt{6^2 - 4(1)(13)}}{2(1)}$$

$$x = \frac{-6 \pm \sqrt{-16}}{2}$$

$$x = -3 \pm 2i$$

Practice Exercises

Practice 1

Solve using the quadratic formula.

$$x^2 - 5x + 6 = 0$$

Practice 2

Solve using the quadratic formula.

$$2x^2 + 7x + 3 = 0$$

Practice 3

Solve using the quadratic formula.

$$3x^2 - 2x - 8 = 0$$

Practice 4

Solve using the quadratic formula.

$$5x^2 + x - 6 = 0$$

Practice 5

Solve using the quadratic formula.

$$4x^2 - 4x + 1 = 0$$

Practice 6

Solve using the quadratic formula.

$$x^2 + 6x + 13 = 0$$

Practice 7

Solve using the quadratic formula.

$$6x^2 - 5x - 6 = 0$$

Practice 8

Solve using the quadratic formula.

$$2x^2 - 8x + 5 = 0$$

Practice 9

Solve using the quadratic formula.

$$7x^2 + 2x - 3 = 0$$

Practice 10

Solve using the quadratic formula.

$$3x^2 + 5x + 1 = 0$$

Activities

Activity 1

Solve using the quadratic formula.

$$x^2 - 7x + 12 = 0$$

Activity 2

Solve using the quadratic formula.

$$2x^2 + 9x + 4 = 0$$

Activity 3

Solve using the quadratic formula.

$$4x^2 - 3x - 10 = 0$$

Activity 4

Solve using the quadratic formula.

$$3x^2 + 2x - 8 = 0$$

Activity 5

Solve using the quadratic formula.

$$9x^2 - 12x + 4 = 0$$

Activity 6

Solve using the quadratic formula.

$$x^2 + 8x + 20 = 0$$

Activity 7

Solve using the quadratic formula.

$$5x^2 - 7x - 6 = 0$$

Activity 8

Solve using the quadratic formula.

$$3x^2 - 10x + 6 = 0$$

Activity 9

Solve using the quadratic formula.

$$8x^2 + x - 2 = 0$$

Activity 10

Solve using the quadratic formula.

$$2x^2 + 7x + 2 = 0$$

Thank you!